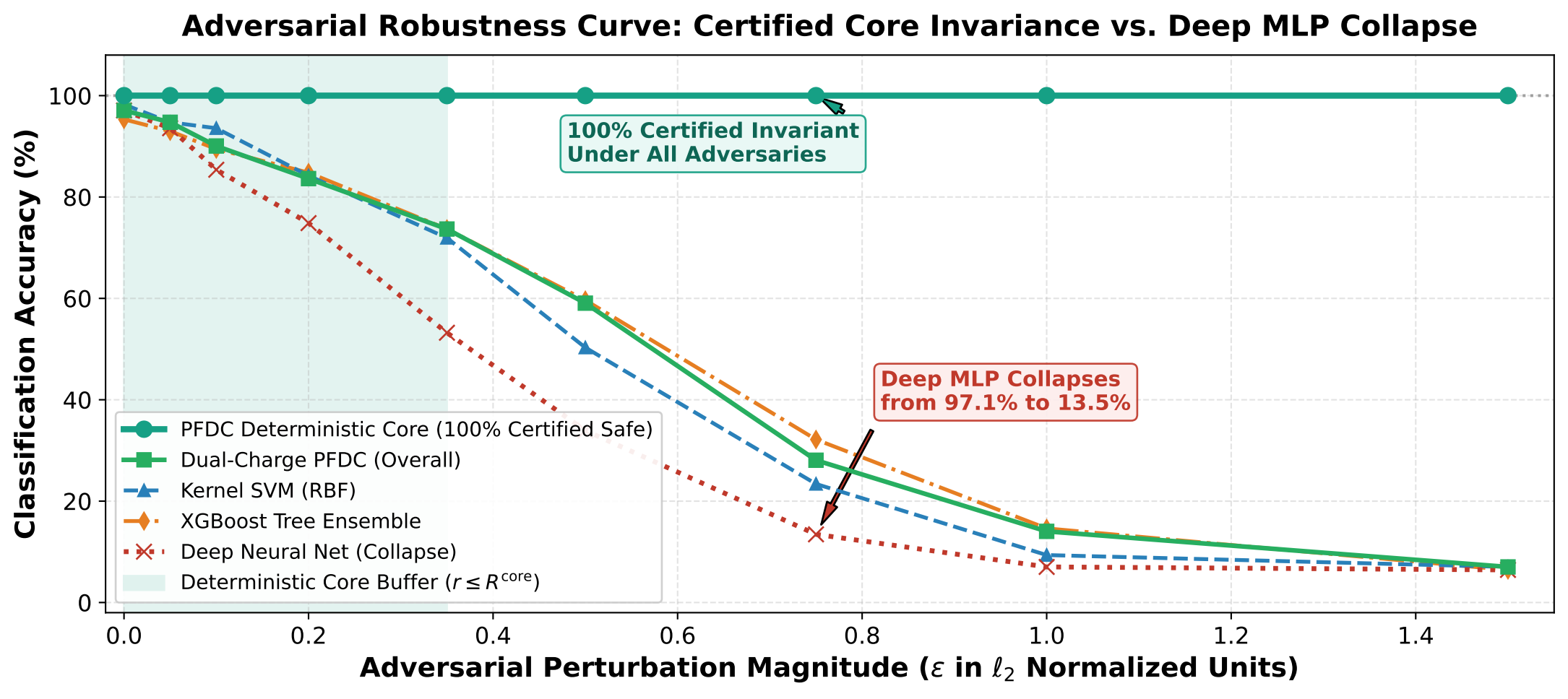
Adversarial Robustness Curve: Certified Core Invariance vs. Deep MLP Collapse
Classification Accuracy (%)
Adversarial Perturbation Magnitude (ε in ℓ2 Normalized Units)
100% Certified Invariant
Under All Adversaries
Deep MLP Collapses
from 97.1% to 13.5%
PFDC Deterministic Core (100% Certified Safe)
Dual-Charge PFDC (Overall)
Kernel SVM (RBF)
XGBoost Tree Ensemble
Deep Neural Net (Collapse)
Deterministic Core Buffer (r ≤ R^core)
0
20
40
60
80
100
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4